

Matplotlib (Python Library) – One Page Notes

What is Matplotlib?

Matplotlib is a Python library used for **data visualization**. It helps create graphs, charts, and plots to understand data easily.

Why Use Matplotlib?

- Visualize data clearly
 - Analyze trends and patterns
 - Create professional charts
 - Works well with NumPy and Pandas
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Installation

```
pip install matplotlib
```

Import Matplotlib

```
import matplotlib.pyplot as plt
```

Basic Line Plot

```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]  
y = [10, 20, 30, 40, 50]
```

```
plt.plot(x, y)  
plt.show()
```

Common Plot Types

1. Line Chart

```
plt.plot(x, y)  
plt.show()
```

2. Bar Chart

```
plt.bar(x, y)  
plt.show()
```

3. Histogram

```
plt.hist(y)  
plt.show()
```

4. Scatter Plot

```
plt.scatter(x, y)  
plt.show()
```

5. Pie Chart

```
plt.pie([30, 40, 30], labels=['A', 'B', 'C'])  
plt.show()
```

Labels and Title

```
plt.plot(x, y)
```

```
plt.title("Sales Report")  
plt.xlabel("Months")  
plt.ylabel("Sales")
```

```
plt.show()
```

Grid

```
plt.grid(True)
```

Change Line Style

```
plt.plot(x, y,  
         color='red',  
         linestyle='--',  
         marker='o')
```

Multiple Lines

```
x = [1,2,3,4]
```

```
y1 = [10,20,30,40]
```

```
y2 = [5,15,25,35]
```

```
plt.plot(x, y1, label="Product A")
```

```
plt.plot(x, y2, label="Product B")
```

```
plt.legend()
```

```
plt.show()
```

Save Figure

```
plt.savefig("graph.png")
```

Matplotlib with NumPy

```
import numpy as np
```

```
x = np.arange(0, 10, 1)
```

```
y = x ** 2
```

```
plt.plot(x, y)
```

```
plt.show()
```

Important Functions

Function	Purpose
<code>plt.plot()</code>	Line chart
<code>plt.bar()</code>	Bar chart
<code>plt.hist()</code>	Histogram
<code>plt.scatter()</code>	Scatter plot
<code>plt.pie()</code>	Pie chart
<code>plt.title()</code>	Add title
<code>plt.xlabel()</code>	X-axis label
<code>plt.ylabel()</code>	Y-axis label
<code>plt.legend()</code>	Show legend
<code>plt.grid()</code>	Show grid
<code>plt.savefig()</code>	Save chart
<code>plt.show()</code>	Display chart

Matplotlib Interview Questions with Answers

1. What is Matplotlib?

Answer:

Matplotlib is a Python library used for creating graphs, charts, and data visualizations. It helps users represent data visually for better analysis and understanding.

Example:

```
import matplotlib.pyplot as plt
```

```
plt.plot([1, 2, 3], [10, 20, 30])
```

```
plt.show()
```

2. What is pyplot?

Answer:

`pyplot` is a module in Matplotlib that provides functions for creating and displaying plots easily. It works similarly to MATLAB plotting commands.

Example:

```
import matplotlib.pyplot as plt
```

```
plt.plot([1, 2, 3], [4, 5, 6])
```

```
plt.show()
```

3. Difference between Bar Chart and Histogram?

Bar Chart	Histogram
Used for categorical data	Used for continuous numerical data

Bars have spaces between them

Bars touch each other

Compares categories

Shows data distribution

```
plt.bar()
```

```
plt.hist()
```

Bar Chart Example:

```
plt.bar(['A', 'B', 'C'], [10, 20, 15])
```

```
plt.show()
```

Histogram Example:

```
plt.hist([10, 20, 20, 30, 40])
```

```
plt.show()
```

4. What is a Scatter Plot?

Answer:

A scatter plot displays individual data points on a graph. It is used to show the relationship between two variables.

Example:

```
plt.scatter([1, 2, 3, 4], [10, 20, 25, 30])
```

```
plt.show()
```

5. How do you add a title to a graph?

Answer:

Use the `title()` function.

Example:

```
plt.plot([1, 2, 3], [10, 20, 30])  
  
plt.title("Sales Report")  
  
plt.show()
```

6. How do you save a plot?**Answer:**

Use the `savefig()` function.

Example:

```
plt.plot([1, 2, 3], [10, 20, 30])  
  
plt.savefig("sales.png")
```

7. How do you display multiple lines on one graph?**Answer:**

Call `plot()` multiple times before `show()`.

Example:

```
x = [1, 2, 3, 4]  
  
plt.plot(x, [10, 20, 30, 40], label="Product A")  
plt.plot(x, [15, 25, 35, 45], label="Product B")  
  
plt.legend()  
  
plt.show()
```

8. What is the use of `legend()`?

Answer:

`legend()` displays labels for different lines or plots, making it easier to identify data series.

Example:

```
plt.plot([1,2,3], [10,20,30], label="Sales")
```

```
plt.legend()
```

```
plt.show()
```

9. What is the use of `grid()`?

Answer:

`grid()` adds grid lines to a graph, improving readability and data interpretation.

Example:

```
plt.plot([1,2,3], [10,20,30])
```

```
plt.grid(True)
```

```
plt.show()
```

10. Why is Matplotlib used with NumPy and Pandas?

Answer:

- **NumPy** provides numerical arrays and mathematical operations.
- **Pandas** provides data handling and analysis.
- **Matplotlib** visualizes the data from NumPy and Pandas.

Example with Pandas:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
data = pd.DataFrame({  
    'Month': ['Jan', 'Feb', 'Mar'],  
    'Sales': [100, 150, 200]  
})  
  
plt.plot(data['Month'], data['Sales'])  
  
plt.show()
```

Short Interview Answers (1-Line)

1. **What is Matplotlib?**
A Python library for data visualization.
2. **What is pyplot?**
A Matplotlib module used to create plots.
3. **Bar Chart vs Histogram?**
Bar Chart compares categories; Histogram shows data distribution.
4. **What is a Scatter Plot?**
A graph showing the relationship between two variables.
5. **How to add a title?**
`plt.title("Title")`
6. **How to save a plot?**
`plt.savefig("file.png")`
7. **How to display multiple lines?**
Use multiple `plt.plot()` functions.
8. **Use of legend()?**
Displays labels for plots.
9. **Use of grid()?**
Adds grid lines to the graph.
10. **Why use Matplotlib with NumPy and Pandas?**
To visualize numerical and tabular data easily.